Day 7 Study Notes

Thu 19 Nov 2026 · Sprint 60 · JEE Main 2027

Part	Time	What to do
Study	3 h 30 min	Physics: Nuclei (75 min) · Maths: Determinants (part 1) (75 min) · Chemistry: Chemical bonding (part 1) (60 min)
Exam	1 h 15 min	The Day 7 exam in the Sprint 60 app: 30 questions, timed, no notes.
Review	1 h 15 min	Open every wrong and skipped question in the app, read its solution, tag the reason, then use Retry my mistakes.

How to use each chapter note. Read it once without stopping. Cover the formulas and write them from memory. Work the solved examples on paper before reading the working. Then read the NCERT section listed at the end.

Physics · Nuclei

About one question per paper every two papers. The syllabus covers nuclear size, mass defect, binding energy, fission and fusion. Radioactive decay is not included.

1. Composition and size

- A nucleus ${}^{A}_{Z}X$ has Z protons and $A - Z$ neutrons.
- Isotopes have the same Z. Isobars have the same A. Isotones have the same number of neutrons.
- Radius: $R = R_0 A^{1/3}$, with $R_0 = 1.2$ fm $= 1.2 \times 10^{-15}$ m.
- Volume is proportional to A, so nuclear density is the same for all nuclei, about $2.3 \times 10^{17}\ \mathrm{kg/m^3}$.

2. Mass and energy

- $E = mc^2$. One atomic mass unit is equivalent to 931.5 MeV.
- **Mass defect:** $\Delta m = Z m_p + (A - Z) m_n - M$, where M is the mass of the nucleus.
- **Binding energy:** $E_b = \Delta m\, c^2 = \Delta m(\text{in u}) \times 931.5$ MeV. It is the energy needed to separate the nucleus into free nucleons.
- Binding energy per nucleon, $\dfrac{E_b}{A}$, measures how stable a nucleus is.

> **Solved example 1.** Find the binding energy of the deuteron. $m_p = 1.0073$ u, $m_n = 1.0087$ u, $m_d = 2.0136$ u.
> $\Delta m = 1.0073 + 1.0087 - 2.0136 = 0.0024$ u. $E_b = 0.0024 \times 931.5 = 2.24$ MeV.

3. The binding energy curve

- E_b/A rises quickly for light nuclei, with peaks at ^{4}He, ^{12}C and ^{16}O.
- It is nearly constant, about 8 MeV, for $30 < A < 170$. This shows the nuclear force is short-range and saturates.
- The maximum is about 8.75 MeV, at ^{56}Fe. For uranium it has fallen to about 7.6 MeV.
- Both very light and very heavy nuclei are less tightly bound than those in the middle. This is why energy is released when light nuclei fuse and when heavy nuclei split.

4. Energy released in a reaction

$$Q = (\text{mass of reactants} - \text{mass of products})c^2 = (\text{total binding energy of products}) - (\text{total binding energy of reactants})$$

- **Fission:** a heavy nucleus splits into two medium ones. One fission of ^{235}U releases about 200 MeV.
- **Fusion:** light nuclei combine. In the Sun, four protons end up as one helium nucleus, releasing about 26.7 MeV. Fusion needs very high temperature to overcome the repulsion between nuclei.

> **Solved example 2.** A nucleus with $A = 240$ and $E_b/A = 7.6$ MeV splits into two fragments with $A = 120$ and $E_b/A = 8.5$ MeV. Find the energy released.
> Binding energy before $= 240 \times 7.6 = 1824$ MeV. After $= 240 \times 8.5 = 2040$ MeV. Energy released $= 2040 - 1824 = 216$ MeV.

5. The nuclear force

- The strongest force in nature, but it acts only over a few femtometres.
- It is the same between proton–proton, neutron–neutron and proton–neutron: it does not depend on charge.
- It is attractive at about 1 fm and becomes repulsive below about 0.8 fm.

Traps. Radius goes as $A^{1/3}$, volume as A, density as A^0. For energy released, multiply binding energy per nucleon by the number of nucleons on each side before subtracting. A higher binding energy per nucleon means more stable, and the products of a reaction that releases energy have more binding energy than the reactants.

Read in NCERT: Physics Class 12 Part 2, chapter "Nuclei": the sections on size, mass–energy, binding energy, nuclear force, fission and fusion.

Mathematics · Determinants (part 1)

Today covers evaluating determinants and their properties. Tomorrow uses them for the adjoint, the inverse and systems of equations.

1. Evaluating a determinant

- Order 2: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$.
- Order 3, expanding along the first row: $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$.
- You may expand along any row or column. Pick the one with the most zeros.
- The determinant of a triangular or diagonal matrix is the product of its diagonal entries.

2. Minors and cofactors

- The minor M_{ij} is the determinant left after deleting row i and column j.
- The cofactor is $C_{ij} = (-1)^{i+j} M_{ij}$.
- The determinant equals the sum of the entries of any row, each multiplied by its own cofactor.
- Entries of one row multiplied by the cofactors of a different row add up to zero.

3. Properties

- $|A^T| = |A|$.
- Swapping two rows (or columns) changes the sign.
- Two identical or proportional rows give zero.
- A common factor of one row can be taken outside. So for an $n \times n$ matrix, $|kA| = k^n |A|$.
- Adding a multiple of one row to another does not change the value.
- $|AB| = |A||B|$, and so $|A^n| = |A|^n$.
- A skew-symmetric matrix of odd order has determinant zero.

A standard result worth remembering: $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$.

4. Area of a triangle

For vertices (x_1, y_1), (x_2, y_2), (x_3, y_3):

$$\text{Area} = \frac{1}{2} \left| \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \right|$$

The three points are collinear when this determinant is zero. When the area is given, remember that the determinant can be $+2 \times \text{area}$ or $-2 \times \text{area}$, which usually gives two answers.

> **Solved example 1.** Find k if the triangle with vertices $(k, 0)$, $(4, 0)$, $(0, 2)$ has area 4.
> Determinant $= k(0-2) - 0 + 1(8-0) = 8 - 2k$. So $\frac{1}{2}|8 - 2k| = 4$, giving $8 - 2k = \pm 8$ and $k = 0$ or $k = 8$.

> **Solved example 2.** A and B are 3×3 matrices with $|A| = 3$ and $|B| = -2$. Find $|2AB|$.
> $|2AB| = 2^3 |A||B| = 8 \times 3 \times (-2) = -48$.

> **Traps.** $|kA|$ is $k^n|A|$, not $k|A|$. The sign pattern of cofactors is a chessboard starting with plus in the top-left corner. In "area is given" questions, do not drop the negative case.

Read in NCERT: Mathematics Class 12 Part 1, chapter "Determinants": determinant of a matrix, area of a triangle, minors and cofactors.

Chemistry · Chemical bonding (part 1)

Chemical bonding gives one to two questions per paper. Today covers shapes, dipole moments and bond character; tomorrow covers hybridisation and molecular orbital theory.

1. The octet rule and its exceptions

Exception	Examples
Incomplete octet	LiCl, $BeCl_2$, BF_3, BCl_3
Odd number of electrons	NO, NO_2
Expanded octet	PF_5, SF_6, H_2SO_4

Formal charge $= V - L - \dfrac{B}{2}$, where V is the number of valence electrons of the free atom, L the number of lone-pair electrons on it, and B the number of bonding electrons around it. In ozone the central oxygen is $6 - 2 - \dfrac{6}{2} = +1$.

2. Ionic bonds and covalent character

- An ionic bond is favoured by low ionisation enthalpy of the metal, a highly negative electron gain enthalpy of the non-metal, and high lattice enthalpy.
- Lattice enthalpy increases with higher ionic charges and smaller ions.
- **Fajans' rules:** covalent character increases with a smaller cation, a larger anion, higher charges, and a cation with a pseudo noble gas (18-electron) configuration.
- So LiCl > NaCl > KCl in covalent character, AlI_3 > $AlCl_3$ > AlF_3, and CuCl > NaCl.

3. Bond parameters and resonance

- Higher bond order means a shorter and stronger bond.
- When a species has several equivalent structures, the real structure is their hybrid and every bond is the same. Bond order $= \dfrac{\text{total number of bonds}}{\text{number of positions}}$.
- CO_3^{2-} and NO_3^-: $\dfrac{4}{3} = 1.33$. O_3 and benzene: 1.5. SO_4^{2-}: 1.5.

4. Dipole moment

- $\mu = q \times d$, measured in debye. It is a vector, and the molecular dipole is the vector sum of the bond dipoles.
- Symmetrical molecules have zero dipole moment even with polar bonds: CO_2, BeF_2, BF_3, CH_4, CCl_4.
- NH_3 (1.47 D) has a larger dipole moment than NF_3 (0.23 D). In NH_3 the lone pair adds to the bond moments; in NF_3 it opposes them.
- CH_3Cl > CH_2Cl_2 > $CHCl_3$ > CCl_4 (zero).

5. VSEPR theory: predicting shape

Electron pairs around the central atom stay as far apart as possible. Repulsion decreases in the order lone pair–lone pair > lone pair–bond pair > bond pair–bond pair.

Bond pairs	Lone pairs	Shape	Examples
2	0	linear	$BeCl_2$, CO_2, HCN
3	0	trigonal planar	BF_3, SO_3
2	1	bent	SO_2, O_3
4	0	tetrahedral	CH_4, NH_4^+
3	1	trigonal pyramidal	NH_3 (107°)
2	2	bent	H_2O (104.5°)
5	0	trigonal bipyramidal	PCl_5
4	1	see-saw	SF_4
3	2	T-shaped	ClF_3
2	3	linear	XeF_2, I_3^-
6	0	octahedral	SF_6

5	1	square pyramidal	BrF_5
4	2	square planar	XeF_4

Counting lone pairs on the central atom. Lone pairs $= \dfrac{\text{valence electrons of central atom} - \text{electrons used in bonds} \pm \text{charge}}{2}$. For XeF_4: $\dfrac{8-4}{2} = 2$. For SF_4: $\dfrac{6-4}{2} = 1$. For I_3^-: $\dfrac{7-2+1}{2} = 3$. In a trigonal bipyramid, lone pairs go in the equatorial positions.

Traps. Shape describes the positions of atoms only; lone pairs are counted to find the geometry but not named in the shape. A molecule with polar bonds can still have zero dipole moment. Double bonds count as one electron group in VSEPR.

Read in NCERT: Chemistry Class 11 Part 1, chapter "Chemical Bonding and Molecular Structure": from the start up to and including VSEPR theory.